

# **Lecture Notes**

## **EC226 Term 1 2025/26**

### **Breakdown of CLRM assumptions**

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# 1 CLRM Assumptions and OLS

To date we have looked at interpretation and estimation for the multiple regression model (1):

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad (1)$$

where the CLRM assumptions are:

1.  $E(\varepsilon_i | \mathbf{x}) = 0$
2.  $Var(\varepsilon_i | \mathbf{x}) = \sigma^2$
3.  $cov(\varepsilon_i, \varepsilon_j | \mathbf{x}) = 0 \quad i \neq j$
4.  $\varepsilon_i | \mathbf{x} \sim N(0, \sigma^2)$

If these 4 assumptions hold then the application OLS to equation (1) yields BLUE estimators:

- Best (smallest variance amongst all unbiased estimators)
- Linear (is a linear function of the  $\varepsilon$ 's)
- Unbiased ( $E(\hat{\beta}_j | \mathbf{x}) = \beta_j$ )
- Estimator

## 2    Breakdown of CLRM Assumptions)

From this point onward we will look at each of these four assumptions and investigate:

1. The implications on the use of OLS in equation (1) if any one of the CLRM assumptions are violated
2. How to test the error term for a violation of each of the assumptions, under the null hypothesis that the assumption is correct.
3. Alternative estimators to OLS to correct for the violation of the assumption.

We are going to investigate the effects on the OLS estimator when:

1. One of the variables,  $\mathbf{x}$ , is not non-stochastic, (see Handout 5).
2. The coefficients in equation (1) are not constant for the sample of observations over which we wish to estimate the model  $i = 1, \dots, n$  (see Handout 6).
3. The coefficients in equation (1) are not constant for some new observations over above the original estimation sample,  $i = n + 1, n + 2 \dots$ , (see Handout 6).
4. Equation (1) has excluded some relevant variables, (see Handout 7).
5. Equation (1) assumes a linear relationship between  $y_i$  and  $\mathbf{x}$  and it should be, for example, quadratic, (see Handout 7).
6. Equation (1) has included some variables that may not actually be that important, (see Handout 7).
7. Non-normality of the error term, (see Handout 8).
8. The variance of the error term in equation (1) is non-constant over the sample,  $V(\varepsilon_i|\mathbf{x}) \neq \sigma^2 \quad \forall i$ , (see Handout 9).
9. One of the variables in  $\mathbf{x}$  or  $y_i$  is measured with error, (see Handout 10).
10. Collinearity between the explanatory variables, (see Handout 11).